

# AJAY GOWDA

Phoenix, AZ, USA

☎ +1 (206) 420-9438 ✉ [ajaysgowda@gmail.com](mailto:ajaysgowda@gmail.com)  [ajaysgowda](https://www.linkedin.com/in/ajaysgowda)

## WORK EXPERIENCE

### REPUBLIC SERVICES

Phoenix, USA

*Sr. Manager, Operations Technology*

Sept 2023 – Present

- Led the team that built and piloted Republic's AI camera based driver safety coaching program.
- Piloted the first fully integrated prototype electric vehicle by training and driving adoption in the field.
- Led the team that focused on continuous improvement and reporting of the AI refuse contamination system.
- Developed and designed billing process for AI refuse contamination detection system that accelerated project by 6 months bringing in \$40 million in revenue.
- Acted as the technical liaison between Republic's business team and IT team.

*Manager, Technical Operations*

Jun 2022 – Aug 2023

- Partnered with Oshkosh Corp to develop the first fully integrated electric refuse vehicle by setting customer requirements for an improved cab and safety features.
- Design and developed the electric vehicle viability models that dictated EV deployment locations.
- Supported the fleet electrification.
- Ran electric vehicle pilots at different location and compiled result to better inform fleet electrification strategy.

### EVBOX

Amsterdam, NL

*Systems Validation Engineer*

Sep 2020 – May 2022

- Design, develop and execute automated firmware tests on current and next gen products for regression testing.
- Develop, deploy and maintain data acquisition infrastructure like databases, visualization tools and vpns on AWS.
- Design and develop one off test setups for unique test scenarios including data acquisition and generating reports.
- Design and develop test benches for correct and repeatable hardware and software validation and verification.
- Assist with pre-compliance testing for certifications such as Energy Star and UL.
- Develop and maintain EOL (End of Line) test applications.

### ETERNAL SUN SPIRE

The Hague, NL

*Development and Validation Engineer*

Aug 2018 – Jul 2020

- Troubleshoot issues with prototype and production Photovoltaic Test Equipment in Lab and Manufacturing lines.
- Design and develop control architecture for complex machines to enable optimal performance and data acquisition.

### GENERAL MOTORS

Milford, USA

*Hybrid Powertrain Development Engineer*

Jan 2016 – Aug 2018

- Develop efficient powertrains with optimal fuel economy and exceptional performance via work on batteries, motors, hydraulic systems, internal combustion engines, and power electronics.
- Analyze and collect data from vehicles on test tracks, in extreme locations, and state of the art high feature test cell using vehicle monitoring software and hardware.
- Validate new hybrid and electric powertrains subjectively and objectively.
- Ensure vehicle test fleet is up to date with latest hardware and software.
- Construct one-off wiring harnesses that include high voltage and CAN communication loops.

*Algorithm Development Engineer – Thermal Controls*

Jul 2015 – Jan 2016

- Developed thermal control system for liquid cooled High Voltage battery systems.
- Software functionality test using SIL and HIL testing techniques.

### UNIVERSITY OF WASHINGTON

Seattle, USA

*Research Assistant at EcoCAR Laboratory*

Oct 2013 – Jun 2015

- Research Assistant at UW EcoCAR Lab.
- Worked on control logic of the supervisory controller that interlinks the two powertrains. (Electric and Diesel)

## EDUCATION

### UNIVERSITY OF WASHINGTON

Seattle, USA

*Master of Science in ME, Control Systems - CGPA - 3.6/4.0*

Sept 2013 – May 2015

- Linear Controls, Digital controls, Optimal Controls
- Master's Thesis - System Level Modeling and Simulation of a Hybrid Vehicle.

### MANIPAL INSTITUTE OF TECHNOLOGY

Manipal, India

*Bachelor of Science in Mechatronics Engineering - CGPA - 8.9/10*

June 2009 – May 2013

- Control Theory, Analog and digital circuit design, Embedded Systems, PLC, Hydraulic and Pneumatic systems.

## TECHNICAL SKILLS

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**Languages:** Python, MATLAB/ Simulink, JAVA, C/C++, TypeScript, HTML, Structured Text (PLC), YAML, XML, SQL  
**Technologies:** AWS, Grafana, Influxdb, Wireguard, MQTT, REST, Docker, Linux, SCADA, MODBUS, OPC, ARM, GIT, Polaron  
**Frameworks:** FastAPI, Django, pytest, opencv  
**Automotive industry tools:** MDA, INCA, Vehicle Spy, dSPACE, neoVI, High Performance Driver Education from GM